

DHEERAJ SRIDHAR REDDY

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EDUCATION

University of Southern California

Los Angeles, CA

Master of Science in Computer Science | GPA: 3.71/4.00

Jan 2024 - Dec 2025

Relevant Coursework: Algorithms, Databases, Machine Learning, Natural Language Processing, Deep Learning, Information Retrieval

PES University

Bengaluru, India

Bachelor of Technology in Computer Science | GPA: 9.21/10

Aug 2019 - May 2023

Specialization: Machine Intelligence & Data Science | Merit Scholarship: Top 20%

Relevant Coursework: Data Analytics, Big Data, Cloud Computing

WORK EXPERIENCE

Scry AI

San Jose, CA

Software Engineer

Jun 2025 - Present

- Enhanced Collatio backend in Node.js with role-based authentication and web-app-token APIs, improving internal API efficiency by 30%, and migrated transactional emails from AWS SES/Mailgun to SMTP, reducing delivery failures by 20%
- Designed and deployed a secure end-to-end parser management system with Angular UI, ClamAV malware scanning, and sandboxed Python execution, reducing parser deployment errors by 40%
- Developed LLM-based keyword, topic, and summary extraction APIs, processing 100+ multi-page documents via dual RQ worker queues, and evaluated multiple open-source models using ROUGE, token similarity, and Jaccard metrics, reducing per-document review time by 35%

Netcore Unbx

Bengaluru, India

Software Engineer

Jul 2023 - Nov 2023

- Engineered a distributed backend system on a Kubernetes kind cluster using Flask and PostgreSQL, supporting 50+ requests/sec with 20% lower latency for e-commerce clients
- Automated configuration backups for the recommendation and search infrastructure in AWS, improving system recovery time by 30% and ensuring data integrity during deployments
- Improved real-time consistency in the recommendation system by implementing NATS-based event messaging and dynamic field filters, reducing subscriber delivery delays by 25% and eliminating 100% of API-level data inconsistencies in MySQL queries

Viga Entertainment Technology

Bengaluru, India

Machine Learning Engineer Intern

Feb 2023 - Jun 2023

- Optimised face capture by enhancing a solver using Image Processing, increasing facial recognition accuracy by 18% and decreasing processing time by 25%
- Leveraged Large Language Models and Stable Diffusion in Movie Colab to automate movie script breakdown and asset extraction, accelerating preproduction processes by 35% and minimizing manual effort by 40 hours per project

PROJECTS

PocketLLM

- Built a full-stack AI assistant (MERN + Ollama) with real-time streaming, JWT-based RBAC, and document-grounded RAG; engineered multi-tier caching, reducing queries by 70% and achieving <5ms latency, fully containerized for local CPU deployment

Multimodal Misinformation Detection

- Engineered a state-of-the-art misinformation detection framework, leveraging TriModal-BridgeTower (BERT + ViT + GraphSAGE), attention-based fusion, and imbalance-handling techniques, boosting accuracy on unseen real-world data from 56 % to 82 %

Sarcasm Detection in Low Resource Indic Languages

- Improved macro F1 score by 18%, from 0.77 with mBERT and IndicBERT to 0.91 using XLM-RoBERTa LLM on 23,000 tweets, handling class imbalance with RandomOverSampler

SKILLS

Programming Languages and Libraries: Python, C, Java, JavaScript, TypeScript, Scikit-Learn, OpenCV, TensorFlow, PyTorch, Stable Diffusion, pandas, NumPy, Keras, Matplotlib, Seaborn, Tableau, Git

Technologies: React.js, Angular, HTML, CSS, REST APIs, Express.js, Node.js, Linux, Flask, LLMs, Kubernetes, Docker

Databases and Cloud Services: PostgreSQL, AWS, Google Cloud Platform, MongoDB, SQL, Excel, NoSQL

PUBLICATIONS

Emotion Recognition Using Text with Emojis and Speech, *International Journal of Advances in Electronics and Computer Science - IJAECs Volume-10, Issue-6 (Jun 2023)*